



CATHOLIC JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

Higher 2

BIOLOGY

Paper 4 Practical

9744/04
26 August 2021

2 hours 30 minutes

Candidates answer on the Question Paper

Additional Materials:

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.
You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
TOTAL	

Answer **all** questions

Before you proceed, read carefully through **the whole** of Questions 1 and 2.

Plan the use of your **time** to make sure that you finish all the work that you would like to do.

You will **gain marks** for recording your results according to the instructions.

- 1** During the light dependent stage of photosynthesis, hydrogen ions and electrons are transferred to hydrogen acceptor molecules, including NADP.

DCPIP (2,6-dichlorophenolindophenol) is a blue dye, which acts as a hydrogen ion and electron acceptor. As DCPIP accepts hydrogen ions or electrons it is reduced and becomes colourless.

You are required to investigate the effect of different colours of light on the rate of the light dependent stage of photosynthesis in a leaf extract containing chloroplasts.

You are provided with:

- a leaf extract in buffered solution, labelled **L**, in a beaker with ice,
- DCPIP solution, labelled **D**,
- filters that allow light of specific colours to pass through, as shown in Table 1.1.

Table 1.1

colour filter	label
purple	U
green	G
orange	O
red	R

The leaf extract is an irritant. It is recommended that you wear safety goggles and gloves.

Proceed as follows.

- 1** Stir the leaf extract, **L**, using the glass rod.
- 2** Use a syringe to draw up 0.5 cm³ of **L**.
- 3** Wipe the outside of the syringe to remove any liquid.
- 4** Put the syringe at the centre of a white tile. This will be used as a colour standard.
- 5** You are required to add 0.2 cm³ DCPIP solution, **D**, to change the colour of the remaining leaf extract, **L**. The change in colour must be sufficient to be observable in the 0.5 cm³ sample transferred to a syringe in step **8**. If the change in colour is not observable, add one or two more drops of DCPIP solution. You are advised not to add more than 0.5 cm³ of DCPIP solution in total.

- 6 Immediately wrap the specimen tube containing the mixture of **L** and **D** in foil. Cover the specimen tube with a foil lid, as shown in Fig. 1.1. This should be easy to remove to obtain the mixture of **L** and **D**. Put back the covered specimen tube containing the mixture of **L** and **D** in the styrofoam cup with ice water.

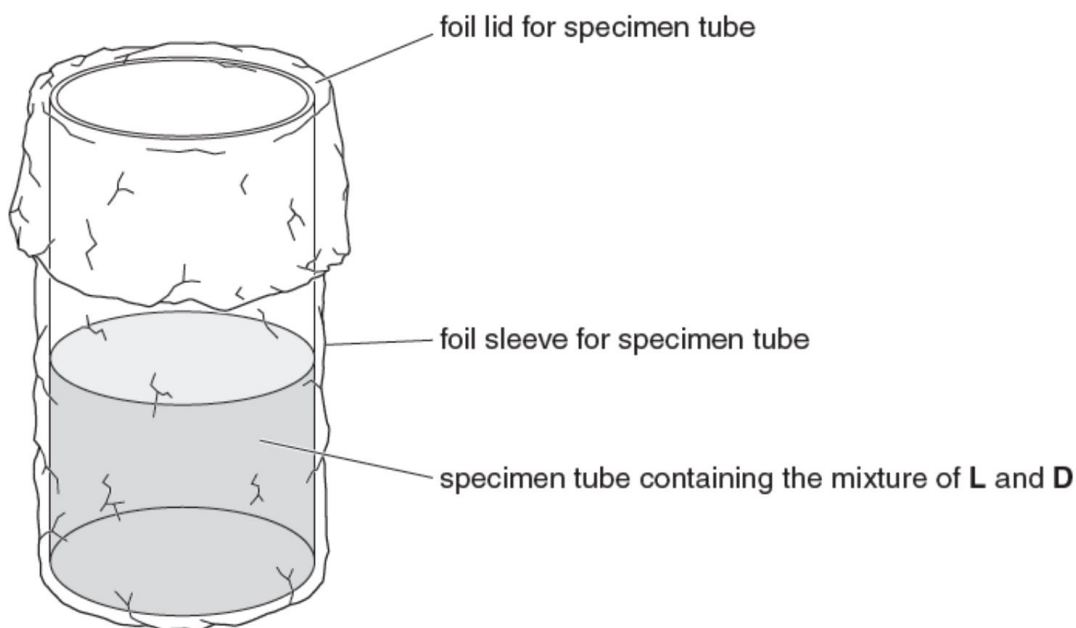


Fig. 1.1

- 7 Place the bench lamp at about 10 cm from the syringe on the white tile. Do **not** switch the lamp on.

The next steps have to be carried out very quickly one after another, so read steps 8 to 15 and refer to Fig. 1.2 before proceeding.

- 8 Remove the foil lid and use a clean syringe to draw up 0.5 cm³ of the mixture of **L** and **D** in the specimen tube. Replace the foil lid immediately.
- 9 Wipe the outside of the syringe and place it next to the colour standard on the white tile. This is the test syringe.
- 10 Immediately cover both syringes with the purple filter, **U**, as shown in Fig. 1.2.
- 11 Switch on the bench lamp and immediately start a stopwatch.

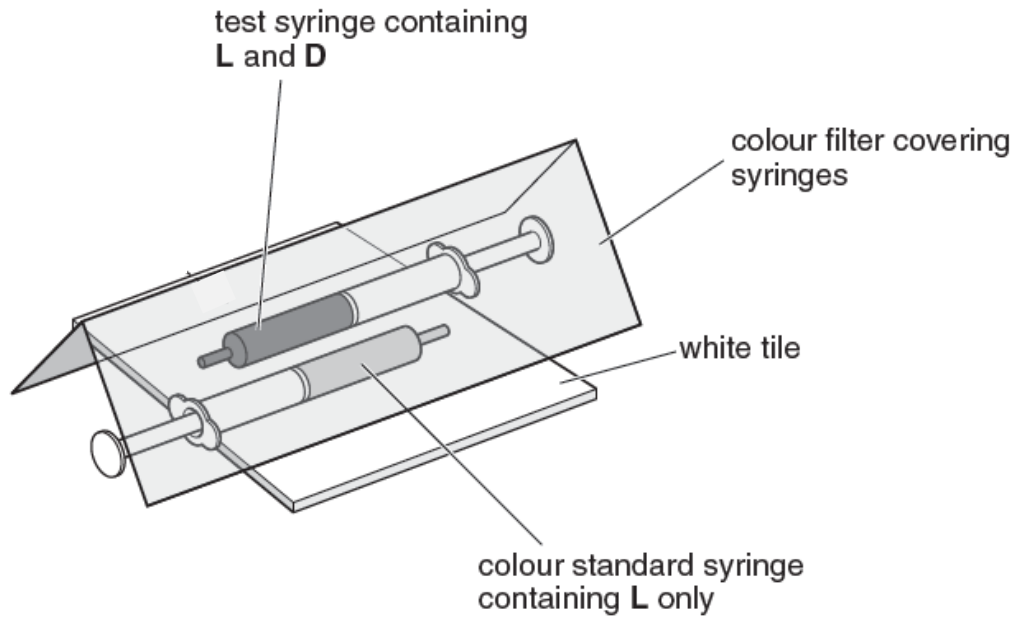


Fig. 1.2

- 12** Measure the time taken for the colour in the test syringe to match that of the colour standard. If the colour does not match after 300 seconds, then record 'more than 300'. Record your results in **(a)(i)**.
 - 13** Switch off the bench lamp.
 - 14** Expel the contents of the test syringe into the beaker labelled **waste**. Rinse the syringe.
 - 15** Repeat steps 8 to 14 using each of the three remaining colour filters (**G**, **O**, and **R**) in turn.
- (a) (i)** Use the space to record, in a suitable format, your results from step **12**.

- (ii) Explain why the leaf extract was kept in ice water.

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..... [1]

- (iii) Explain why the leaf extract containing DCPIP was kept covered by foil.

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..... [1]

- (iv) Describe a suitable control that could have been set up for this investigation. Justify your answer.

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..... [2]

- (v) Explain why rate of photosynthesis is different with the different colour filters.

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..... [2]

- (vi) The results of this investigation may be limited by lack of replication and sources of error.

Describe **one other** significant source of error in the procedure and suggest an improvement that can be made.

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..... [2]

- (b) In another similar investigation, a student collected leaves from two varieties of the same species of a garden plant that has different coloured leaves.

Variety **A** dark red leaves
 Variety **B** green and white striped leaves

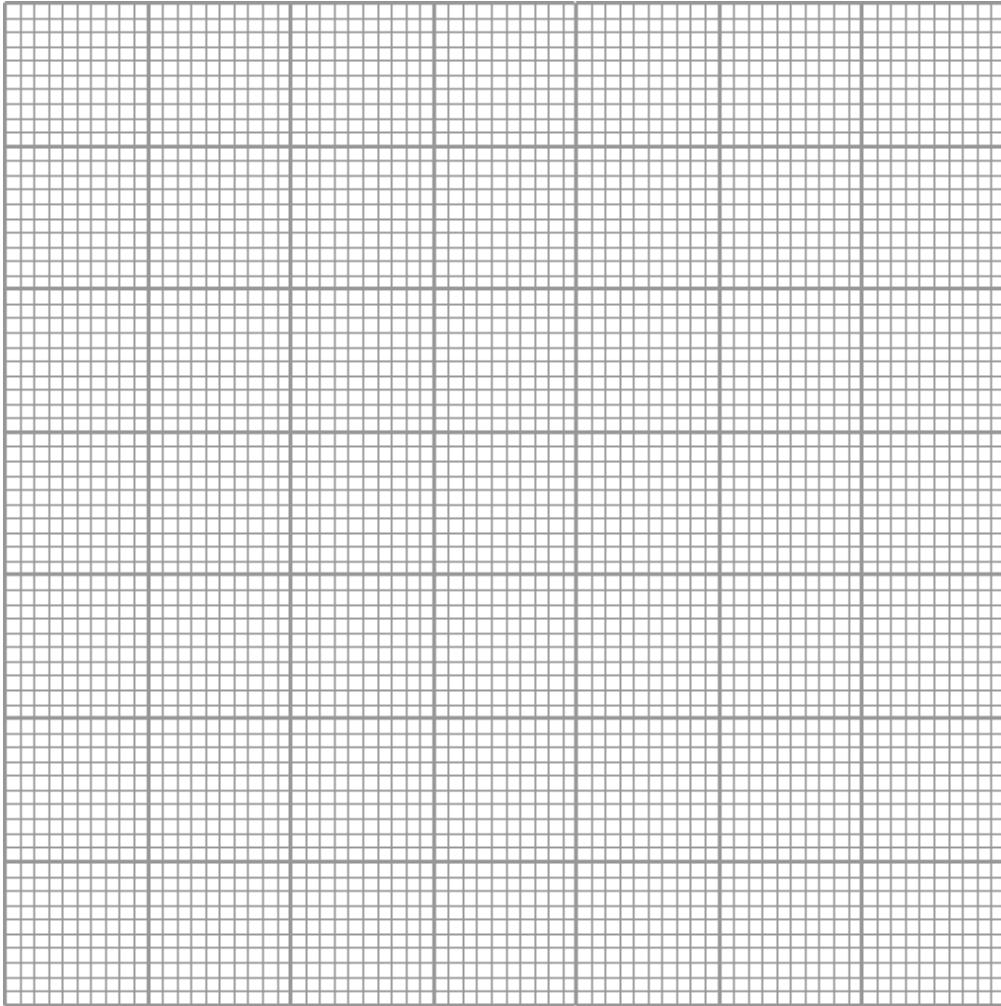
The student made a chloroplast extract from the leaves of each variety and measured the rate of photosynthesis for each extract in different wavelengths of light.

Table 1.2 shows the rates of photosynthesis calculated by the student from her results.

Table 1.2

wavelength of light / nm	rate of photosynthesis / s ⁻¹	
	dark red leaf	green and white striped leaf
425	0.097	0.081
450	0.071	0.063
525	0.023	0.023
625	0.030	0.039
675	0.057	0.058

- (i) Use the grid provided to display the results shown in Table 1.2 in appropriate form.



[4]

- (ii) Use the graph in (i) to estimate the rate of photosynthesis at a wavelength of 430 nm in plants with dark red leaves.

..... [1]

(c) A student deduced that DCPIP can be used to measure the amount of ascorbic acid (vitamin C) present in fruit juice **J**. He found a bottle of solution labelled '1% ascorbic acid' in the school laboratory.

- (i) Add 1.0 cm³ of 1% ascorbic acid into a test-tube. Using a syringe, add DCPIP solution dropwise into the test-tube until in excess. Record your observations.

.....

..... [1]

- (ii) Use the information provided to plan a method that you can use to estimate the concentration of ascorbic acid in fruit juice **J**. Do **not** plan to carry out repeats.

You will carry out this method in (c)(iii).

In your plan, you must use:

- 1% ascorbic acid
- DCPIP

You may select from any of the other apparatus and materials provided for **Q1**.

Your planned method should:

- have a clear and helpful structure so that the method described could be repeated by anyone reading it
- include details to ensure that results are as accurate and repeatable as possible
- use the correct technical and scientific terms
- only make use of the apparatus and materials provided.

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(iii) Carry out your method described in (c)(ii) to estimate the concentration of ascorbic acid in fruit juice J.

(iv) Use the space below to record, in a suitable format, your results from (c)(iii).

[3]

(v) State the concentration of the ascorbic acid in fruit juice J.

Concentration of ascorbic acid in fruit juice J%
[1]

[Total: 27]

- 2 During this question you will require access to a microscope and slide **E1**.

An eyepiece graticule scale is attached to your microscope.

Slide **E1** is a microscope slide of a stained transverse section through a leaf of a land plant.

You are not expected to be familiar with this specimen.

- (a) Observe slide **E1** using the 10X objective.

You are required to:

- use the eyepiece graticule to measure the depth of the midrib **Y** of the leaf as shown in Fig 2.1 and the depth of the vascular bundle at the midrib;
- draw a plan diagram of the leaf.

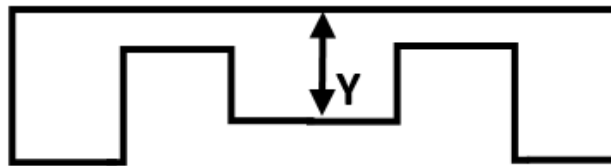


Fig 2.1

- (i) Use the eyepiece graticule in the microscope to measure:

- the depth of the leaf at the mid-rib **Y**:
..... eyepiece graticule units
- the depth of the vascular bundle at the mid-rib :
..... eyepiece graticule units

Use these measurements to determine the simplest whole number ratio of the depth of the leaf at mid-rib **Y** to the depth of the vascular bundle at the mid-rib.

Show all the steps in your working.

simplest whole number ratio
[2]

- (ii) Outline how you can use the method described in **(a)(i)** to help you draw an accurate plan diagram of the leaf section in slide **E1** showing the arrangement and correct proportions (and shape) of the different tissues without calibrating the eyepiece graticule scale.

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..... [2]

- (iii) Draw a large plan diagram of the leaf section in slide **E1**. No cells should be drawn. Label one vascular bundle.

[4]

- (b)** You are provided with a plastic ruler.

Use the plastic ruler to calibrate your eyepiece graticule under 100X magnification.

- (i)** Explain why this is a less accurate method compared to the use of a stage micrometer.

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..... [2]

- (ii)** Calculate the magnification of your plan drawing in **a(iii)**.
Show your working clearly.

[2]

- (iii) Select a group of cells consisting of **two** adjacent palisade mesophyll cells that are touching **two** adjacent epidermal cells in slide **E1**.

Make a detailed drawing of this group of four cells and label any internal structures you can see in one of the palisade mesophyll cell.

[4]

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- (c) Fig 2.2 (on **page 17**) shows a photomicrograph of a transverse section of a leaf obtained from a plant that grows under a different environmental condition. Several structural differences are seen compared to that in slide **E1**.

Prepare the space below so that it is suitable for you to record these differences, other than size, between the leaves in slide **E1** and Fig. 2.2.

Record your observations in the space that you have prepared.

[4]

- (d) "Slide **E1** is from a xerophytic plant that grows well in areas with low water availability."
State two visible features in the leaf specimen in slide **E1** that support the statement above.

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..... [2]

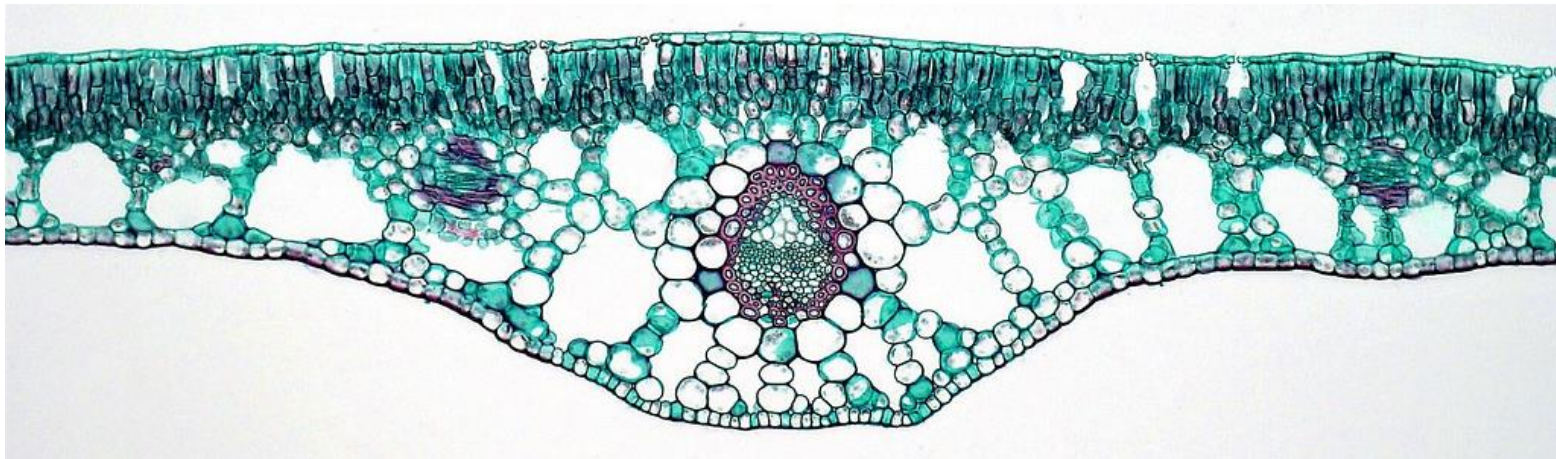


Fig 2.2

(e) You are required to:

- find the area of the field of view, using Fig. 2.3 shown below;
- count the number of stomata in a quarter of the field of view, using Fig. 2.4 (on **page 19**);
- calculate the mean number of stomata per mm^2 .

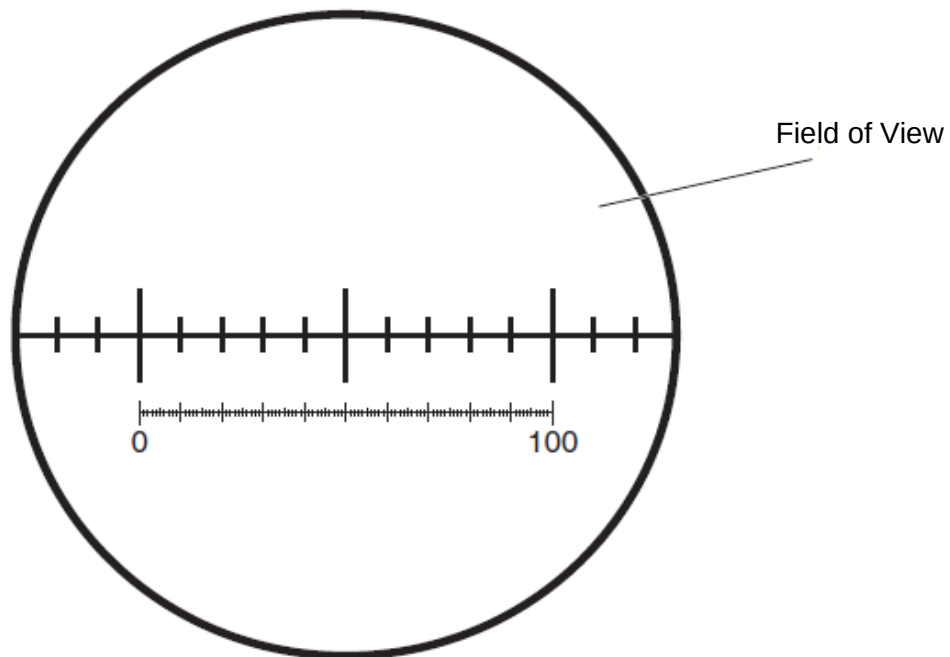


Fig 2.3

The smallest measurement on this stage micrometer is 0.1 mm.

- (i) Calculate the area of the field of view (FOV), using the formula for the area of a circle: πr^2 where $\pi = 3.14$ and r = radius of the FOV.

Show your working and use appropriate units.

area of field of view (FOV)..... mm^2
[2]

Fig. 2.4 is a photomicrograph of the stomata in a field of view of the lower epidermis of a leaf from another plant. It was observed under the same magnification as in Fig. 2.3, hence under a field of view of the same size.

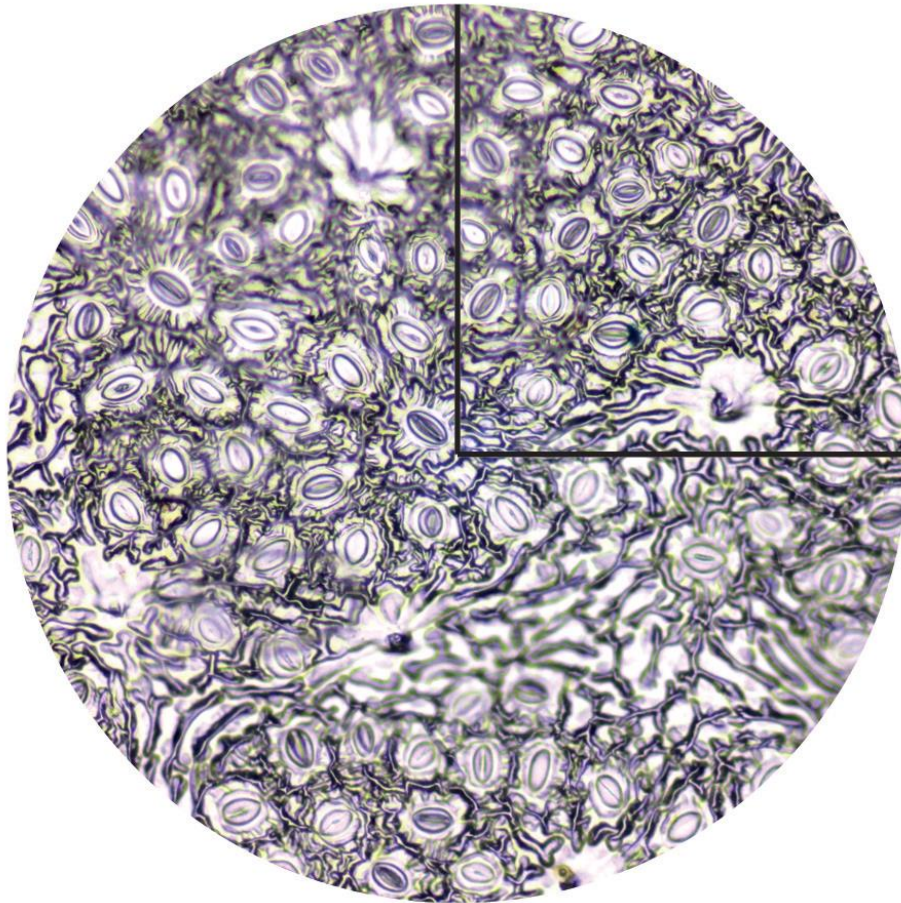


Fig. 2.4

There are too many stomata to count so the use of a sampling method is recommended to estimate the number of stomata in this field of view.

- (ii) Count and record a sample number of whole stomata in the quarter of the field of view shown in Fig. 2.4.

Mark clearly with an 'X' on each of the stomata counted in Fig. 2.4

Number of stomata in the quarter of the field of view:
[1]

- (iii) Based on the methods used in **e(i)** and **e(ii)**, describe clearly the procedure used to estimate the mean number of stomata per mm^2 of this plant.

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..... [3]

[Total: 28]

END-OF-PAPER